

CPSC 375 High-Performance Computing

Lecture 7

Parallel Hardware

Flynn's Taxonomy

classic von Neumann

SISD Single instruction stream Single data stream	SIMD Single instruction stream Multiple data stream
MISD Multiple instruction stream Single data stream	MIMD Multiple instruction stream Multiple data stream

not covered

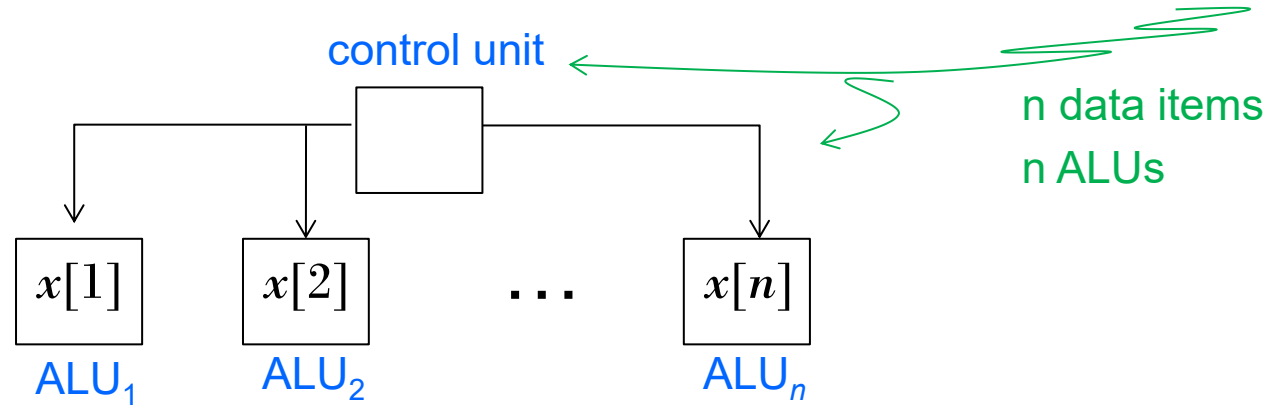
SISD

- Traditional sequential computer (von Neumann architecture).
- A single processor executes one instruction stream to operate on a single data stream.
- Instructions are executed one after another in a linear fashion.
- Older, single-core desktop PCs or microcontrollers.

SIMD

- Designed for *data-level parallelism*, where the same operation is performed on multiple data points simultaneously.
- A single control unit broadcasts one instruction to multiple processing elements, each working on its own data.
- Ideal for vector processing, matrix operations, and image processing.
- Modern GPUs or CPU instruction sets like Intel's AVX (Advanced Vector Extensions).

SIMD example



```
for (i=0; i<n; i++)  
    x[i] += y[i];
```

SIMD

- What if we don't have as many ALUs as data items?
 - Divide the work and process iteratively.
- e.g., with 4 ALUs and 15 data items:

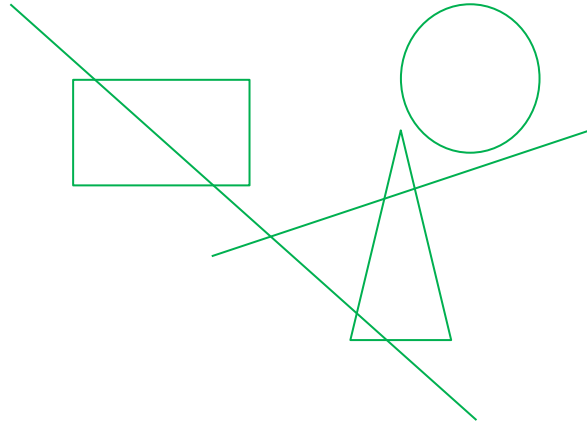
Round3	ALU ₁	ALU ₂	ALU ₃	ALU ₄
1	X[0]	X[1]	X[2]	X[3]
2	X[4]	X[5]	X[6]	X[7]
3	X[8]	X[9]	X[10]	X[11]
4	X[12]	X[13]	X[14]	

SIMD drawbacks

- All ALUs are required to execute the same instruction, or remain idle.
- In classic design, they must also operate synchronously.
- Efficient for large data parallel problems, but not other types of more complex parallel problems.

Graphics Processing Units (GPU)

- Real time graphics API's use points, lines, and triangles to internally represent the surface of an object.



GPUs

- A graphics processing pipeline converts the internal representation into an array of pixels that can be sent to a computer screen.
- Several stages of this pipeline (called **shader functions**) are programmable.
 - Typically just a few lines of C code.



GPUs

- Shader functions are also implicitly parallel, since they can be applied to multiple elements in the graphics stream.
- GPU's can often optimize performance by using SIMD parallelism.
- The current generation of GPU's use SIMD parallelism (General-Purpose GPU).

MISD

- A rare architecture where multiple processors perform different operations on the same piece of data.
- Multiple instruction streams are applied to a single data stream.
- Generally used for fault tolerance or specialized redundant systems.
- Space shuttle flight control computers or specialized digital signal processing.

MIMD

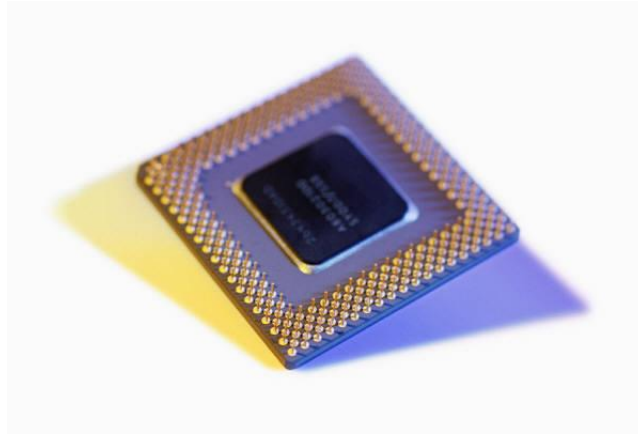
- The most common architecture for modern HPC clusters and multi-core systems.
- Multiple autonomous processors simultaneously execute different instructions on different data.
- *Shared memory vs. distributed memory:*
 - Shared Memory: All processors access a global memory space, often requiring semaphores to prevent race conditions.
 - Distributed Memory: Each processor has its own local memory; coordination is handled via Message Passing Interface (MPI).

Shared Memory System

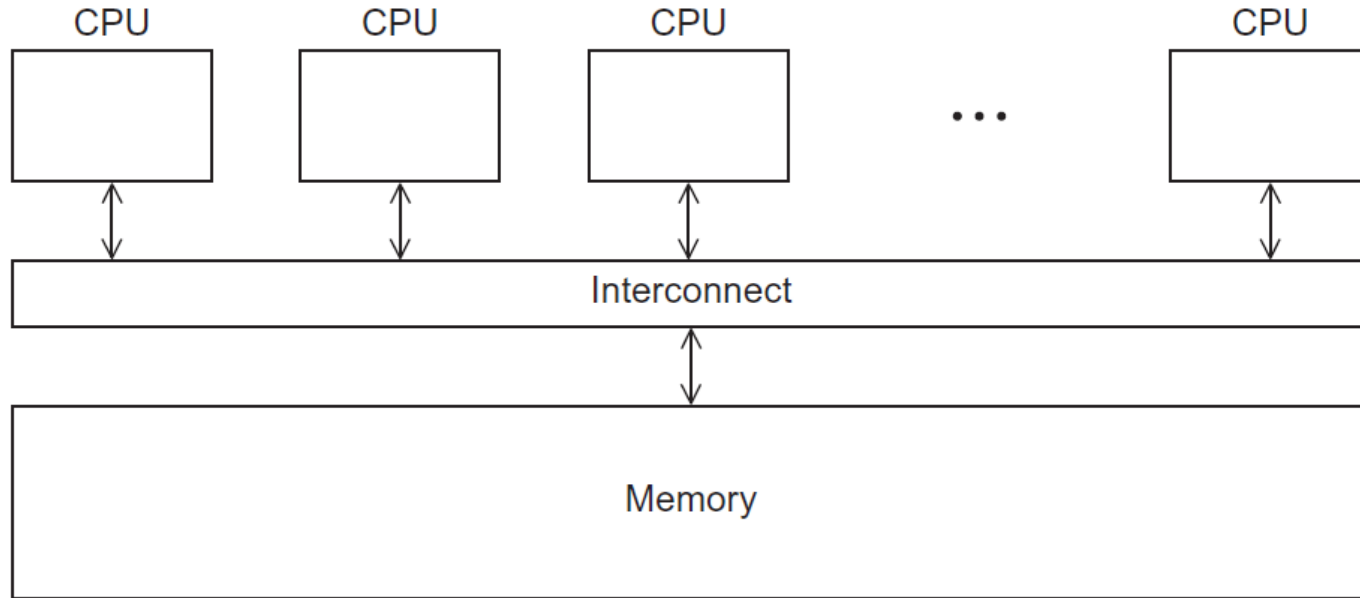
- A collection of autonomous processors is connected to a memory system via an interconnection network.
- Each processor can access each memory location.
- The processors usually communicate implicitly by accessing shared data structures.

Shared Memory System , cont'd

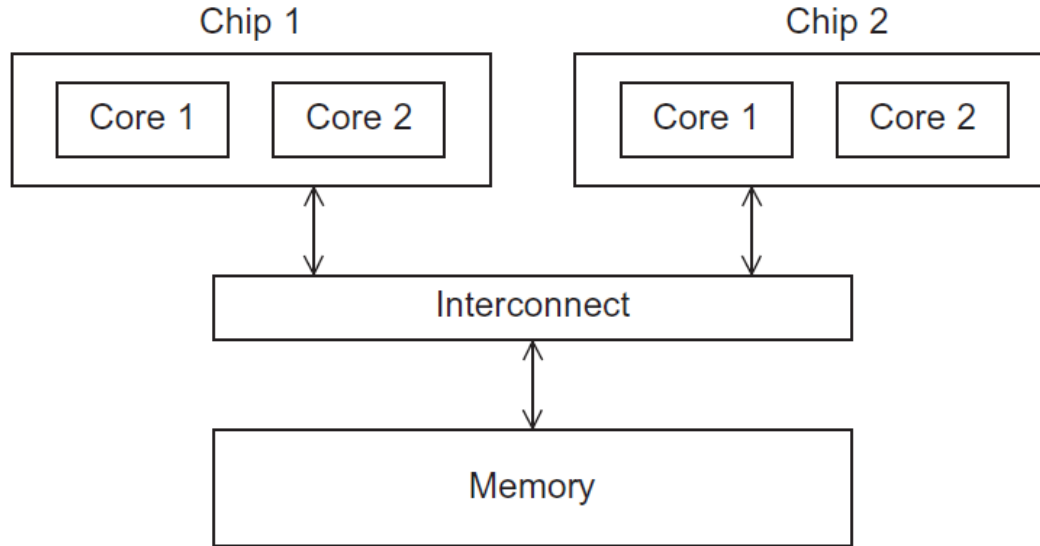
- Most widely available shared memory systems use one or more multicore processors.
 - (multiple CPU's or cores on a single chip)



Shared Memory System , cont'd

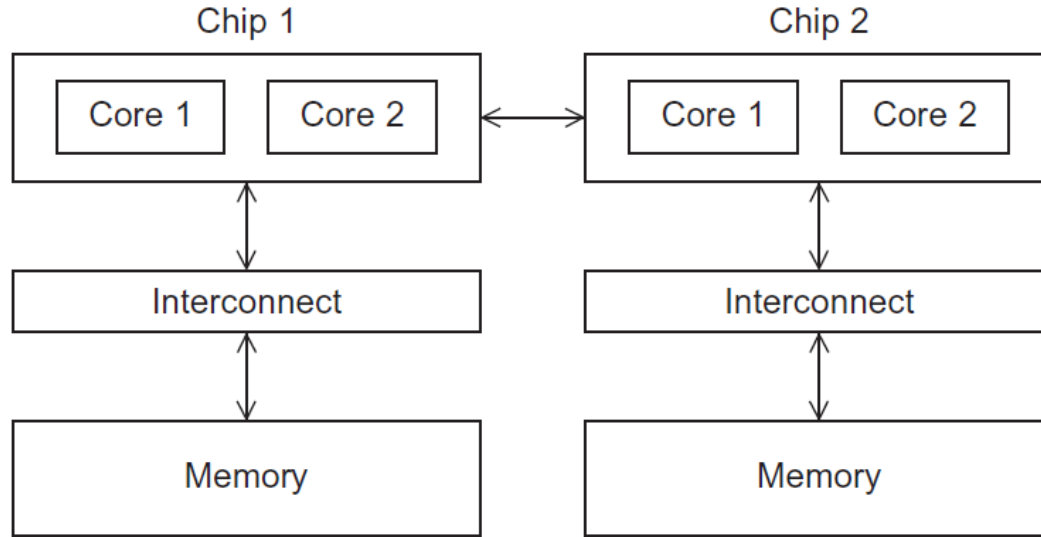


UMA Multicore System



Time to access all the memory locations will be the same for all the cores.

NUMA Multicore System

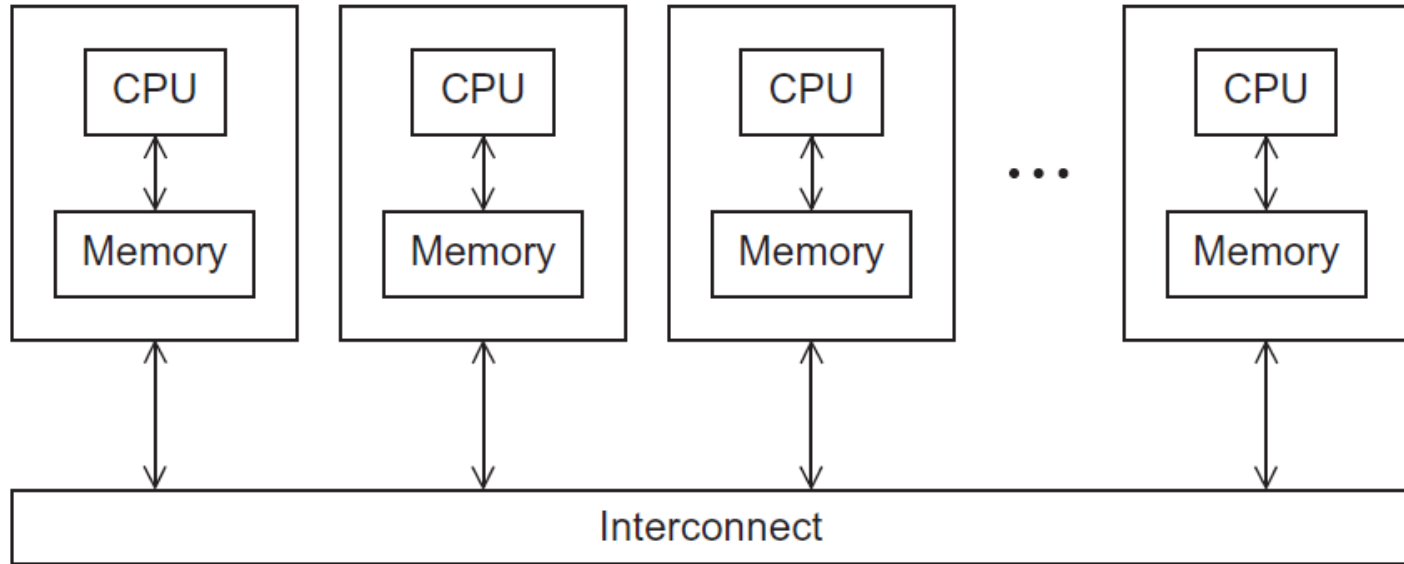


A memory location a core is directly connected to can be accessed faster than a memory location that must be accessed through another chip.

Distributed Memory System

- *Clusters*
 - A collection of commodity systems
 - Connected by a commodity interconnection network
- *Nodes* of a cluster are individual computations units joined by a communication network.

Distributed Memory System





Build Your Own Cluster!

- *Compute nodes*
 - "workhorses" of the cluster
- *Head node (master node, front-end)*
 - login, manage, schedule
- **High-speed interconnect**
 - data communication
- **Network storage**
 - parallel file systems
- **Software stack**
 - MPI, Slurm, BLAS, etc.

